DEVELOPMENT OF AN AI-BASED REAL-TIME ANOMALY DETECTION AND AUTONOMOUS DECISION SUPPORT SYSTEM FOR DEEP SPACE MISSIONS

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THE PROBLEM

- Deep-space missions (Chandrayaan-3, Aditya-L1, Mars Rover) face 20+ min communication delays
- Earth cannot intervene in real time during a fault
- Traditional threshold-based monitoring misses gradual sensor degradation
- AI must detect anomalies, predict failures, and act autonomously

DATASET DETAILS

NASA C-MAPSS-1 TURBOFAN ENGINE DEGRADATION DATASET

- 100 training engines, 100 test engines
- 21 sensor channels per cycle, run-to-failure telemetry
- 14 informative sensors retained after feature selection
- Degraded label: RUL < 30 cycles
- Average engine lifespan: ~206 cycles

THREE-MODEL ANOMALY DETECTION PIPELINE

- Isolation Forest — unsupervised, 100 trees, no labels needed
- Autoencoder — 14→10→5→10→14, flags high reconstruction error
- LSTM Classifier — 30-cycle sliding window, captures temporal degradation trends
- LSTM RUL Regressor — predicts remaining useful life, capped at 125 cycles

MODEL PERFORMANCE

- Isolation Forest: Precision 0.71 | Recall 0.68 | F1 0.69
- Autoencoder: Precision 0.78 | Recall 0.74 | F1 0.76
- LSTM Classifier: Precision 0.85 | Recall 0.82 | F1 0.83 (best)
- RUL Prediction: MAE ~18 cycles | RMSE ~25 cycles
- Fleet: ~52 normal | ~35 warning | ~13 critical engines

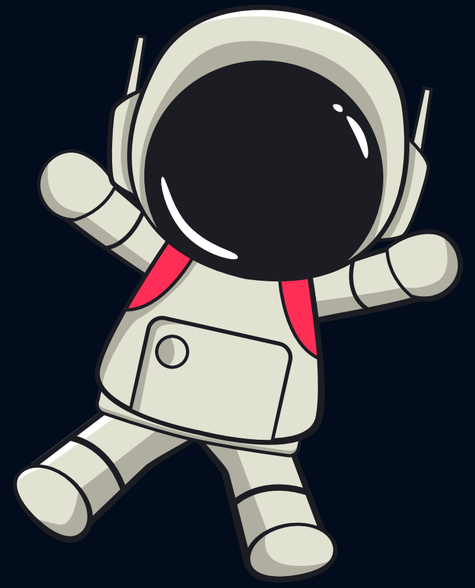
RISK SCORING :

1. 0.00–0.24 → NORMAL — continue operations
2. 0.25–0.44 → MONITOR — send telemetry to ground
3. 0.45–0.64 → WARNING — reduce engine load
4. 0.65–0.79 → HIGH RISK — switch to backup engine
5. 0.80–1.00 → EMERGENCY — shutdown now

MISSION CONTROL DASHBOARD

- Matplotlib FuncAnimation renders live sensor feed cycle-by-cycle
- Anomaly dots appear the moment any model flags a cycle
- RUL countdown gauge updates in real time
- Alert log captures every flagged event with timestamp and action
- All driven by actual trained model outputs, not simulation

FUTURE MISSION IDEAS

- Deploy on edge hardware (TensorFlow Lite, INT8 quantisation) for onboard use
- Extend to CMAPSS FD002–FD004 (6 conditions, multiple fault modes)
- Replace rule-based decisions with Deep Q-Network (reinforcement learning)
- Apply to Aditya-L1 solar wind telemetry for storm detection
- Federated learning across multiple spacecraft without sharing raw data
- Add SHAP explainability so ground engineers can audit every decision

THANK YOU !